Po (Eric) Peng

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Skills

- **Languages**: C/C++, Python, Bash, Verilog
- **Embedded Systems**: embedded Linux, RTOS (µC/OS-II, Zephyr), MCU firmware, ARM, UEFI/EDK II, bootloaders
- **Interfaces & Protocols**: I2C, UART, TCP/IP, Ethernet, CAN, Modbus
- **Development Tools**: Git, Makefile, GitLab CI/CD, GitHub, TRACE32, Valgrind, Coverity, Docker

Work Experience

Qualcomm San Diego, CA

Embedded Software Engineer - Intern 06/2026 – 09/2026

- Developed the **first UEFI prototype** on a new architecture using **C**, integrating platform-specific **UART** support, implementing **timer interrupt** support, and debugging boot flow to reach the UEFI interactive shell
- Contributed to open-sourcing downstream **EDK II** firmware by adapting upstream validation checks for internal **CI/CD** and resolving integration issues

Moxa Taipei, Taiwan

Embedded Software Engineer - R&D 06/2021 – 10/2024

Protocol Gateways (Embedded Linux)

- Led **cross-functional development** of MGate 5192's **reusable Linux-based C protocol stack** and backend, implementing socket communication and packet handling to cut new-product integration from 3 months to 1 month
- Built a proprietary CAN protocol module from scratch and evaluated **multithreaded transmission strategies** across multiple ports, analyzing race conditions, mutex contention, and transaction-scanning overhead
- Improved an MGate 5000 RESTful library using **IPC**, reducing API development time by 20%
- Designed an SD card backup module and resolved firmware issues through **Linux kernel analysis**
- Established automated testing pipelines with Valgrind and Coverity analysis in **GitLab CI/CD**

Media Converters (MCUs)

- Led **MCU firmware development in C** for an Ethernet-to-fiber Converter from kickoff to market launch, coordinating with hardware, product, and SQA teams
- Resolved customer communication issues on a CAN-to-fiber Converter by tuning the CAN FD sample point and correcting **RAM configuration errors** affecting high-speed transmission

Education

University of Washington Expected 03/2027

M.S. Electrical and Computer Engineering | GPA 3.94/4

- Coursework: Parallel Computer Architecture, GPU Parallel Computing, Digital Systems Design with FPGAs

National Taiwan University of Science and Technology | M.S. Electrical Engineering | GPA 3.92/4.3 08/2020

Chang Gung University | B.S. Electrical Engineering | GPA 3.7/4.0 06/2018

Projects

Parallel Computing on a 2,000-Core RISC-V System 01/2026 – 03/2026

- Optimized **C++** kernels through architecture-aware parallelization and latency hiding
- Managed explicit data movement and synchronization without cache coherence, using **cycle-accurate simulation** to identify compute, memory, and NoC bottlenecks

Embedded Sensing Capstone — UW SEAL 01/2026 – 06/2026

- Built an **ESP32** sensing system with on-device signal processing, ML inference, and LoRa communication for industrial gas leak detection

RTOS Implementation 09/2018 – 01/2019

- Modified the **µC/OS-II RTOS** kernel to implement and evaluate scheduling algorithms

Leadership & Mentoring

University of Washington Seattle, WA

FIRST Robotics Mentor 09/2025 – 12/2025

- Mentored students in electronics, sensors, wiring, and software framework design, emphasizing **PID control**, Java programming, and computer vision